

CodeAlpha Internship - Data Science

Task 1

Project Title: Iris Flower Classification using Machine Learning
Domain: Data Science & Machine Learning

1. Project Overview & Objective

The objective of this project is to build an end-to-end classification model using the Iris flower dataset. The dataset includes measurements of sepals and petals for three distinct flower species: Setosa, Versicolor, and Virginica. We train a robust classification model to learn structural boundaries and predict the species of unseen samples with high precision.

2. Complete Python Source Code

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

iris_dataset = load_iris()
feature_dataframe = pd.DataFrame(data=iris_dataset.data,
                                columns=iris_dataset.feature_names)
target_series = pd.Series(iris_dataset.target, name="species")

(training_features, testing_features, training_labels, testing_labels) =
train_test_split(
    feature_dataframe, target_series, test_size=0.2, random_state=42,
    stratify=target_series
)

feature_scaler = StandardScaler()
scaled_training_features = feature_scaler.fit_transform(training_features)
scaled_testing_features = feature_scaler.transform(testing_features)

classification_model = RandomForestClassifier(n_estimators=100, random_state=42)
classification_model.fit(scaled_training_features, training_labels)

model_predictions = classification_model.predict(scaled_testing_features)
accuracy = accuracy_score(testing_labels, model_predictions)

print("Iris Flower Classification Results:")
```

```
print("Accuracy Score:", round(accuracy * 100, 2), "%")
print("\nClassification Report:\n", classification_report(testing_labels,
model_predictions, target_names=iris_dataset.target_names))
print("Confusion Matrix:\n", confusion_matrix(testing_labels, model_predictions))
```

3. Execution Output

Iris Flower Classification Results:

Accuracy Score: 100.0 %

Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	10
virginica	1.00	1.00	1.00	10
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Confusion Matrix:

```
[[10  0  0]
 [ 0 10  0]
 [ 0  0 10]]
```